

COBAR AP, Australia

WMO: 947100

Lat: 31.5389S

Lon: 145.7964E

Elev: 715

StdP: 14.32

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.1	32.5	18.1	14.1	66.8	20.9	16.1	63.4	20.3	60.3	18.3	58.3	0.9	60	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.6	104.3	67.2	101.1	66.5	98.0	65.7	73.5	86.1	71.8	86.1	70.4	86.1	11.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.8	116.7	76.5	68.5	107.4	75.6	66.2	99.2	75.1	37.6	86.1	36.0	84.8	34.7	85.8	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	18.1	16.3	DB	25.7	109.3	2.1	3.8	24.1	112.0	22.9	114.2	21.7	116.3	20.1	119.1	(4)
(5)			WB	24.6	75.5	2.3	2.6	22.9	77.4	21.6	79.0	20.3	80.4	18.6	82.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	82.9	80.1	74.3	65.6	56.5	50.9	49.3	52.2	59.5	66.1	73.1	78.0	(6)
(7)		DBStd	13.30	7.20	7.11	6.87	6.91	6.19	5.44	5.17	6.19	7.18	7.56	7.90	7.18	(7)
(8)		HDD50	186	0	0	0	0	13	53	75	42	3	0	0	0	(8)
(9)		HDD65	1965	0	1	10	76	272	424	485	400	200	79	16	2	(9)
(10)		CDD50	5884	1019	843	752	468	213	79	55	111	287	498	692	867	(10)
(11)		CDD65	2190	554	424	297	94	7	0	0	5	34	113	258	404	(11)
(12)		CDH74	28976	7869	5491	3348	1023	78	2	0	57	531	1538	3419	5620	(12)
(13)		CDH80	15354	4759	3087	1547	316	2	0	0	9	150	623	1739	3122	(13)
(14)	Wind	WSAvg	7.8	9.4	8.8	7.9	6.8	6.0	6.1	6.3	7.0	7.9	8.7	9.1	9.3	(14)
(15)	Precipitation	PrecAvg	16.0	1.7	1.7	1.4	1.0	1.3	1.2	1.1	1.1	1.0	1.4	1.4	1.4	(15)
(16)		PrecMax	30.1	9.2	7.4	8.6	7.9	5.7	4.4	4.0	3.0	4.2	4.1	6.3	6.0	(16)
(17)		PrecMin	4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	6.7	2.1	1.9	1.6	1.5	1.2	1.0	1.0	0.9	1.1	1.0	1.5	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	109.5	107.7	99.5	91.8	79.5	72.6	71.6	80.8	89.1	95.3	104.0	105.4	(19)
(20)			MCWB	68.3	69.9	66.2	62.7	59.5	57.4	54.3	57.7	59.0	61.4	64.6	65.2	(20)
(21)		2%	DB	105.1	102.6	95.4	87.2	76.2	67.9	67.5	74.1	84.1	91.3	98.6	101.2	(21)
(22)			MCWB	67.8	67.7	65.6	62.3	58.2	55.1	53.0	54.5	57.6	60.6	63.9	64.9	(22)
(23)		5%	DB	101.9	98.5	91.9	83.4	73.0	64.7	64.1	69.8	80.1	87.1	94.1	97.7	(23)
(24)			MCWB	67.5	66.6	64.8	60.8	56.8	53.7	51.7	53.0	56.7	59.7	62.8	64.3	(24)
(25)		10%	DB	98.4	94.8	88.5	80.1	69.5	62.1	61.2	66.4	75.9	82.8	89.7	94.2	(25)
(26)			MCWB	66.9	66.6	64.4	59.8	55.8	52.6	50.6	51.4	55.4	58.8	61.8	63.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	75.5	75.0	69.0	65.0	60.6	60.1	60.6	64.1	66.6	70.5	73.3	(27)
(28)			MCD B	87.3	87.0	81.4	78.6	69.3	66.3	65.0	69.8	75.4	79.2	85.0	83.6	(28)
(29)		2%	WB	73.0	73.6	71.5	65.7	61.7	58.2	55.9	57.4	61.6	64.7	68.8	70.4	(29)
(30)			MCD B	88.3	87.1	80.6	75.2	69.0	62.9	61.7	69.1	73.2	78.0	82.4	83.6	(30)
(31)		5%	WB	71.5	71.9	69.4	63.7	59.4	56.2	53.8	54.8	59.4	62.8	67.2	68.7	(31)
(32)			MCD B	89.4	85.9	81.5	76.0	68.8	61.6	60.2	67.0	72.7	77.1	80.8	84.2	(32)
(33)		10%	WB	70.1	70.2	67.4	62.0	57.6	54.3	51.8	52.6	57.2	60.9	65.6	67.0	(33)
(34)			MCD B	89.4	85.3	81.5	76.2	67.1	60.4	59.7	64.4	72.6	78.8	80.9	84.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	24.6	23.1	23.6	24.7	23.6	21.3	22.6	25.4	26.4	26.1	25.3	25.9	(35)
(36)			MCDBR	27.9	27.5	26.7	27.8	26.3	24.2	25.8	28.5	30.6	30.5	30.1	29.4	(36)
(37)			MCWBR	9.3	9.5	10.1	11.7	12.8	13.8	14.4	14.0	13.4	11.8	10.7	10.7	(37)
(38)		5% WB	MCD BR	21.8	20.5	20.1	20.9	18.7	16.4	18.9	24.2	23.1	22.9	21.6	22.4	(38)
(39)			MCWBR	8.7	7.5	8.9	9.5	10.0	10.1	11.1	12.8	11.4	10.3	8.2	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.338	0.317	0.300	0.287	0.283	0.280	0.289	0.323	0.325	0.338	0.330	(40)	
(41)		taud	2.584	2.600	2.642	2.650	2.644	2.648	2.638	2.593	2.481	2.514	2.522	2.566	(41)	
(42)		Ebn at Noon	318	312	308	298	286	280	286	297	300	312	315	321	(42)	
(43)		Edh at Noon	33	32	29	25	23	22	23	26	33	34	35	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2451	2182	1898	1491	1123	902	1021	1322	1753	2112	2307	2489	(44)	
(45)		RadStd	124	150	134	80	59	55	59	79	96	143	193	112	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page